Ming-Qiang Liang

Ph.D. Candidate in Paleoclimate Dynamics
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Research Interests

Paleoclimate modeling; Asian monsoon dynamics; AMOC variability; proxy-model comparison; Heinrich events.

Education

Ph.D. Paleoclimate Dynamics, UCLouvain **(2026)**

Temporal and spatial climate diversity and underlying mechanisms over East Asia during Heinrich Events 1–6 (Advisor: Prof. Qiuzhen Yin)

M.S. Geoscience, National Taiwan University **(2021)**

Paleoclimate evolutions during the Holocene: A multi-proxy stalagmite record from Majiaping Cave, Guizhou, China (Advisor: Prof. Hongchun Li)

B.S. Geoscience, Southwest University **(2018)**

Characteristics and Influencing Factors of Air Environment Variability in Furong Cave, Chongqing (Advisor: Prof. Tingyong Li)

Awards

National Taiwan University Graduate Scholarship (2019–2020)

Outstanding Graduate, Southwest University (2018)

Scholarship of Southwest University (2017)

Funding Awards

Temporal and spatial climate diversity and underlying mechanisms over East Asia during Heinrich Events 1-6. *FNRS doctoral aspirant funding*: **Liang, M.**, €138,900, 01/10/2022-30/09/2026

Publications

† Co-first author; * Corresponding author. Author contribution designations are retained from the confirmed source.

In Preparation

Liang, M., Yin, Q. et al. (2026). CO2 forcing drives divergent tropical hydroclimate during Heinrich stadials.

Published Journal Articles

2026

[1] Wu, Z., Yin, Q., Berger, A., Guo, Z., Zhang, Z., **Liang, M.**, Liu, W. (2026). High- and Low-Latitude Insolation Forcing Drives the Half-Precession Cycle in Winter Arctic Sea Ice. *Geophysical Research Letters*, 53(19), e2026GL125062. https://doi.org/10.1029/2026gl125062

[2] Sun, W.-J., Tao, Z.-P., Zhang, Z.-Y., Lan, T.-Q., Li, T., Zi, P., Zhang, J., Wu, Y., **Liang, M.-Q.**, Chen, C.-J., Mu, Y.-Y., Li, J.-Y., Li, T.-Y. (2026). Coupled influence of Western Pacific moisture anomalies and mid-tropospheric humidity on October precipitation in Yunnan Province Southwest China. *Atmospheric Research*, 336, 108830. https://doi.org/10.1016/j.atmosres.2026.108830

2025

[3] Chen, C.-J., Dong, Y.-X., Wu, Y., Zhang, J., Wang, H.-B., Xue, Y.-X., Sun, W.-J., Xie, M.-P., Cheng, H., Li, J.-Y., **Liang, M.-Q.**[*], Li, T.-Y. (2025). The variation of the East Asian summer monsoon on multi-centennial scale during 8.3–2.9 ka BP. *Global and Planetary Change*, 255, 105115. https://doi.org/10.1016/j.gloplacha.2025.105115

[4] Zhao, J., Cheng, H., Pan, L., Wang, K., Cui, J., **Liang, M.-Q.**, Dong, X., Lu, J., Li, Q., Yang, Y. (2025). Putting the recent 100-year of Asian summer monsoon weakening into context. *Science Bulletin*, 70(18), 2959-2962. https://doi.org/10.1016/j.scib.2025.05.006

[5] Zhang, J., **Liang, M.-Q.**, Li, J.-Y., Wu, Y., Xue, Y.-X., Sun, W.-J., Liu, X.-Z., Chen, J.-K., Chen, C.-J., Li, T.-Y. (2025). Upstream moisture sources and circulation patterns controls Asian monsoon hydroclimate over the past 20,000 years. *Quaternary Science Reviews*, 363, 109438. https://doi.org/10.1016/j.quascirev.2025.109438

[6] Liu, Y., Hu, C., Hu, Z., Liao, J., **Liang, M.**, Yin, Q. (2025). Extended Duration of Abrupt Climate Events From the Early to Late Holocene. *Geophysical Research Letters*, 52(7), e2025GL115543. https://doi.org/10.1029/2025gl115543

2024

[7] **Liang, M.-Q.**, Yin, Q., Sun, Y., Zhang, C., Berger, A., Lyu, A., Liu, W., Wu, Z. (2024). Distinct response of Asian summer monsoon circulation and precipitation to orbital forcing during six Heinrich events. *Quaternary Science Reviews*, 344, 108946. https://doi.org/10.1016/j.quascirev.2024.108946

2023

[8] Zhang, J., Genty, D., Sirieix, C., Verdet, C., Mateo, S., Devaux, L., Sgubin, G., Bujan, S., Minster, B., Régnier, E., Li, T.-Y., Bourges, F., **Liang, M.-Q.**, Chen, C., Marieu, V. (2023). Data-model comparisons of isotopic and hydrological variabilities of the karstic vadose zone above Villars Cave, SW-France based on 20 years' monitoring record. *Chemical Geology*, 642, 121802. https://doi.org/10.1016/j.chemgeo.2023.121802

[9] Xue, Y.-X., Zhang, J., Su, Z., Wu, Y., Liang, Q.-s., **Liang, M.-Q.**, Liu, X.-Z., Chen, C.-J., Qiu, H.-Y., Li, J.-Y., Li, T.-Y. (2023). Quantifying source effects based on rainwater δ18O from 10-year monitoring records in Southwest China. *Applied Geochemistry*, 155, 105706. https://doi.org/10.1016/j.apgeochem.2023.105706

[10] Wu, Y., Li, T.-Y., Li, J.-Y., Cheng, H., Ning, Y.-F., Shen, C.-C., Yang, Y., Zhao, J.-Y., Chen, C.-J., **Liang, M.-Q.**, Xiao, S.-Y., Qiu, H.-Y., Xu, Y.-Z., Huang, Y.-Y., Yu, T.-L., Edwards, R.L. (2023). Variations in Asian summer monsoon and hydroclimate during Heinrich stadials 4 revealed by stalagmite stable isotopes and trace elements. *Quaternary Science Reviews*, 299, 107869. https://doi.org/10.1016/j.quascirev.2022.107869

2022

[11] Wu, Z., Yin, Q., **Liang, M.**, Guo, Z., Shi, F., Lu, H., Su, Q., Lü, A. (2022). The effect of astronomical forcing on water cycle: Sea ice and precipitation. *Chinese Science Bulletin*, 68(12), 1443-1458. https://doi.org/10.1360/tb-2022-0833

[12] Zhang, J., **Liang, M.-Q.**[†], Li, T.-Y., Chen, C.-J., Li, J.-Y. (2022). Asian-Australian monsoon evolution over the last millennium linked to ENSO in composite stalagmite δ18O records. *Quaternary Science Reviews*, 281, 107420. https://doi.org/10.1016/j.quascirev.2022.107420

Preprints and Other Existing Outputs

2021

[1] Chen, C.-J., Yuan, D.-X., Li, J.-Y., Wang, X.-F., Cheng, H., Ning, Y.-F., Edwards, R.L., Wu, Y., Xiao, S.-Y., Xu, Y.-Z., Huang, Y.-Y., Qiu, H.-Y., Zhang, J., **Liang, M.-Q.**, Li, T.-Y. (2021). The 4.2 ka event in East Asian monsoon region, precisely reconstructed by multi-proxies of stalagmite. *Preprint*. https://doi.org/10.5194/cp-2021-20

Conference Presentations

Oral Presentations

[1] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *The 7th PAGES Open Science Meeting*. Shanghai, China. 2025

[2] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *EGU*. Vienna, Austria. 2024

[3] Asynchronous Holocene Optimum in East Asia monsoon region recorded by stalagmites and its underlying climate dynamics. *XXI Congress of the International Union for Quaternary Research (INQUA)*. Rome, Italy. 2023

[4] Paleoclimatic evolutions during the Holocene: A stalagmite δ18O record from Majiaping Cave, Guizhou, China. *EGU*. Vienna, Austria. 2022

[5] The highest resolution stalagmite over Holocene in southwest China by acute 14C dating. *Climate Change: The Karst Record IX*. Innsbruck, Austria. 2021

Poster Presentations

[1] Variations of ENSO linking with the intensity of Asian-Australian monsoon revealed by stalagmite and rainfall δ18O over the past millennium. *The 28th General Assembly of the International Union of Geodesy and Geophysics*. Berlin, Germany. 2023

[2] Chronology construction of a stalagmite from southwest China by 14C, 230Th/U and 210Pb dating methods. *The 8th east Asia accelerator mass spectrometry symposium*. Nagoya, Japan. 2019

Research Experience

Sun Yat-Sen University, Guangzhou, China (2024.10)

Climate Feedback–Response Analysis Method (CFRAM)

Xiamen University, Xiamen, China (2023.02)

Isotope-enabled climate modeling

Institute of Atmospheric Physics, CAS, Beijing, China (2022.12)

Regional climate modeling (RegCM4)

Technical Skills

Climate models: HadCM3, iCESM, RegCM4.7

Programming: Matlab, Fortan, Bash, Slurm

Climate Diagnostics: Data Processing & Analysis, Physical Mechanism Diagnostics, Climate Model Evaluation

Paleoclimate: Proxy, Dating, Uncertainty

Academic Service

Journal reviewer:

Quaternary Science Reviews; Climate of the Past; Climate Dynamics; Global and Planetary Change